

Ring Formation Maneuver: Single-Integrator Kinematics

Dzung Tran*, David Casbeer[†], Eloy Garcia[‡], and Isaac E. Weintraub[§]
U.S. Air Force Research Laboratory, Wright-Patterson Air Force Base, OH 45433

Dejan Milutinović[¶]
University of California Santa Cruz, Santa Cruz, CA 95064

Satyanarayana G. Manyam^{||}
Infoscitex Corporation, Dayton, OH 45431

Most conventional formation control algorithms consider convergence towards a specific desired point (or a specific desired displacement) for each vehicle relative to their neighbors. However, this specification restricts the flexibility in the motion of the vehicles. Motivated from this standpoint, this paper considers a formation control problem, where the desired position of each vehicle in the formation is described by a set of points. In particular, we consider relative motion in a two-vehicle system consisting of a leader and a follower, which is scalable to a larger formation in a distributed manner. The leader is free to move in 3-dimensional space while the follower seeks to converge to the desired set of points. We especially consider the desired set to be a ring of a given radius located at a certain distance from the leader in this paper. In addition, we propose two control algorithms. Both of them allow the follower to alter the set point smoothly during the mission and thus increase the follower's flexibility of motion. The stability of the formation under the proposed control algorithms is analyzed using the Lyapunov theory. Numerical examples are also provided to illustrate the efficacy of the control algorithms.

I. Introduction

Many teamwork behaviors can be observed in nature such as migrating birds flying in V-shaped formation to reduce the drag force and save energy for long trips, ants working together to find and transport food back to their nest or fish swimming in schools not only to increase their sensing capability but also to confuse their predator, and hence boost survival rates [1–3]. Motivated by these benefits of teamwork, a lot of research effort has been made toward developing algorithms for multiagent systems, which consist of a group of robots or vehicles (i.e., agents) communicating and collaborating for achieving a global objective. One branch of multiagent systems is formation control, where control algorithms are designed to allow vehicles to form and maintain the desired formation under adverse conditions and dynamic environment. Several results on optimal relative positions for vehicles in formation flights are investigated in [4–7].

Formation control has a wide range of applications including satellite clusters, search and rescue missions, security patrol, to name but a few examples (see, for example, [8] and references therein). Representative results in the fields are [8–31]. A common denominator of these formation control algorithms is that they define a specific desired point (or a specific desired displacement) for each vehicle relative to their neighbors. This, however, restricts the flexibility of the follower vehicles in space. For instance, when the lead vehicle is agile, it requires the follower vehicles to put a lot of effort to keep up. Therefore, it is crucial to come up with new schemes for allowing flexibility in the followers' movement while the original benefits of teamwork are still guaranteed.

The contribution of this paper is to proposed novel control algorithms for solving a formation control problem, where the desired position of each vehicle in the formation is a set of points. In particular, we consider relative motion in a two-vehicle system consisting of a leader and a follower, which is scalable to a larger formation in a distributed

*Research Associate, Control Science Center of Excellence.

[†]Team Lead, Control Science Center of Excellence, AIAA Associate Fellow.

[‡]Research Engineer, Control Science Center of Excellence, AIAA Member.

[§]Electronics Engineer, Aerospace Systems Directorate, Air Force Research Laboratory, AIAA Senior Member.

[¶]Professor, Department of Electrical and Computer Engineering.

^{||}Research Scientist, Contractor for Air Force Research Laboratory, AIAA Member.

manner. The leader is free to move in 3-dimensional space while the follower seeks to converge to an arbitrary point of the desired set. We especially consider the desired set to be a ring of a given radius located at a certain distance from the leader in this paper. In addition, two control algorithms are proposed. The first one explicit defines a point within the desired set (i.e., the ring). This set point changes depending on the current position of the follower. The second one utilizes the potential field method (see, for example, [11, 12, 32, 33]) as a tool to shape the desired set. Both schemes allow the follower to alter the set point smoothly during the mission and thus increase the follower's flexibility. In addition, the stability of the system is analyzed using the Lyapunov theory.

The organization of this paper is as follows. In Section II, we present the problem formulation. In Sections III and IV, the two different control approaches are respectively proposed along with the stability analysis. Discussion on the implementation and the two methods are presented in Section V. Numerical examples are provided in Section VI to illustrate the efficacy of the proposed methods. Finally, concluding remarks are summarized in Section VII.

II. Problem Formulation

In this paper, we consider a system consisting of two vehicles: a leader and a follower. The leader and the follower are free to move in 3-dimensional Cartesian space. The follower attempts to approach any position on a ring of radius, R , located a distance, h , behind the leader as illustrated in Figure 1. Our objective is to design the controller for the follower to achieve this task. For that purpose, define $p_l \triangleq [p_{lx}, p_{ly}, p_{lz}]^T$ and $p_f \triangleq [p_{fx}, p_{fy}, p_{fz}]^T$ as positions of

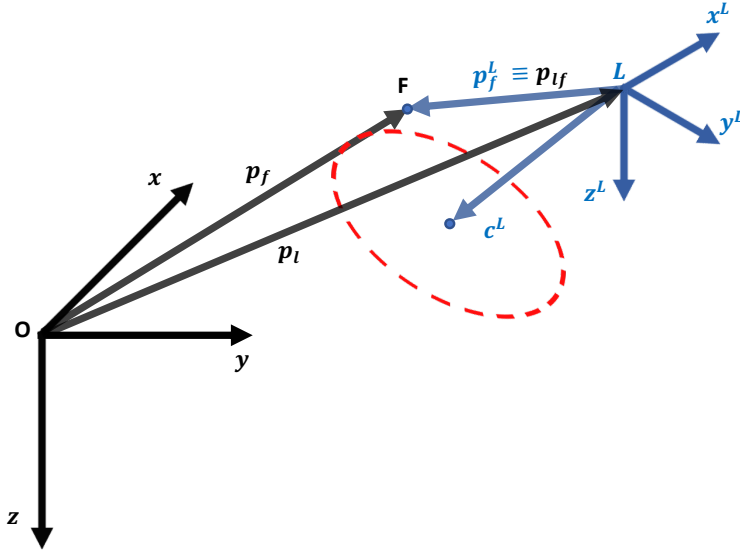


Fig. 1 An illustration of vectors in the inertial frame (black) and the leader body frame (blue) where the leader is located at L , the follower is located at F , and the center of the ring is located at the tip of c^L .

the leader and the follower in the inertial frame, respectively. The single integrator dynamics of the follower is given by

$$\dot{p}_f(t) = u(t), \quad (1)$$

where $u(t) \in \mathbb{R}^3$ is the control input. Here, our goal is to design the controller $u(t)$ so that the follower approaches to the desired ring.

As illustrated in Figure 1, we have

$$\mathcal{R}_i^L(\gamma, \chi)(p_f(t) - p_l(t)) = \mathcal{R}_i^L(\gamma, \chi)p_{lf}(t) = p_f^L(t), \quad (2)$$

or equivalently,

$$p_f(t) = p_l(t) + \mathcal{R}_L^i(\gamma, \chi) p_f^L(t), \quad (3)$$

where $p_{lf}(t)$ denotes the vector $\vec{LF}$ in the inertial frame, $p_f^L(t)$ is the position of the follower in the leader frame (i.e., the vector $\vec{LF}$ in the leader frame), and $\mathcal{R}_L^i(\gamma, \chi)$ is the rotation matrix for the transformation from the leader frame to inertial frame defined by

$$\mathcal{R}_L^i(\gamma, \chi) \triangleq \mathcal{R}_1(\chi) \mathcal{R}_2(\gamma) = \begin{bmatrix} \cos(\chi) & -\sin(\chi) & 0 \\ \sin(\chi) & \cos(\chi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\gamma) & 0 & \sin(\gamma) \\ 0 & 1 & 0 \\ -\sin(\gamma) & 0 & \cos(\gamma) \end{bmatrix}, \quad (4)$$

with γ and χ being the leader's flight path angle and course angle, respectively. In addition, the rotation matrix satisfies the properties $(\mathcal{R}_L^i(\gamma, \chi))^{-1} = (\mathcal{R}_L^i(\gamma, \chi))^T = \mathcal{R}_L^i(\gamma, \chi)$ and $\det(\mathcal{R}_L^i(\gamma, \chi)) = 1$. Next, taking time derivative of (3) yields

$$\dot{p}_f(t) = \dot{p}_l(t) + \mathcal{R}_L^i(\gamma, \chi) \dot{p}_f^L(t) + \dot{\mathcal{R}}_L^i(\gamma, \chi) p_f^L(t), \quad (5)$$

where

$$\dot{\mathcal{R}}_L^i(\gamma, \chi) = \left(\frac{\partial \mathcal{R}_1(\chi)}{\partial \chi} \dot{\chi}(t) \right) \mathcal{R}_2(\gamma) + \mathcal{R}_1(\chi) \left(\frac{\partial \mathcal{R}_2(\gamma)}{\partial \gamma} \dot{\gamma}(t) \right) \quad (6)$$

with $\dot{\chi}(t)$ and $\dot{\gamma}(t)$ being the time rate of change of the leader's course angle and the flight path angle, respectively. As a result, the controller $u(t)$ has the form of the right hand side of (5). While $p_f^L(t)$ can be calculated directly from (2), we are free to design the term $\dot{p}_f^L(t)$. In other words, we now translate the problem to designing the controller for the follower in the leader frame*. For that purpose, let $c^L \triangleq [c_x^L, c_y^L, c_z^L]^T$ be the center of the ring, $r^{*L} \in \mathbb{R}^3$ be an arbitrary vector in the set of vectors depicting the ring in the leader frame, and the single integrator dynamics of the follower in the leader frame is given by

$$\dot{p}_f^L(t) = u^L(t). \quad (7)$$

The objective now is to design the controller $u^L(t)$ such that at steady state

$$p_f^L(t) = r^{*L}. \quad (8)$$

In the following sections, we propose two methods for solving the problem in the leader frame: the first one explicitly defines the desired vector depicting the ring while the second one utilizes objective functions to construct potential functions and the potential fields.

III. The Explicit Method

A. Proposed Control Algorithm

In this section, we aim to design the controller in the leader frame by explicitly defining the vector r^{*L} . Specifically, we consider

$$r^{*L} \triangleq \begin{bmatrix} c_x^L \\ R \cos(\varphi) + c_y^L \\ R \sin(\varphi) + c_z^L \end{bmatrix}, \quad (9)$$

with R being the radius of the ring and $\varphi(t) \in \mathbb{R}$ being updated under the dynamics

$$\dot{\varphi}(t) = \dot{p}_{fy}^L(t) \sin(\varphi) - \dot{p}_{fz}^L(t) \cos(\varphi), \quad \varphi(0) = \varphi_0, \quad (10)$$

*Note that one can design the single integrator controller in the body frame for calculating the term $\dot{p}_f^L(t)$, then substitute it back into (5) to obtain the controller $u(t)$ for implementation on the follower in the inertial frame.

where $\dot{p}_{fy}^L(t)$ and $\dot{p}_{fz}^L(t)$ are the second and third components of $\dot{p}_f^L(t)$. Note that for any specific value of $\varphi \in (-\pi, \pi]$, $\varphi + 2n\pi$ for all $n \in \mathbb{Z}$ represents the same point on the desired ring. We are now able to define the error

$$e^L(t) \triangleq \begin{bmatrix} e_x^L(t) \\ e_y^L(t) \\ e_z^L(t) \end{bmatrix} = p_f^L(t) - r^{*L} \in \mathbb{R}^3. \quad (11)$$

Next, the proposed control algorithm for the follower in the leader frame is given by

$$u^L(t) = -\alpha e^L(t), \quad (12)$$

where α is a positive constant gain.

B. Stability Analysis

For stability analysis, taking the time derivative of (11) yields

$$\dot{e}^L(t) = \dot{p}_f^L(t) - \dot{r}^{*L}, \quad e^L(0) = e_0^L. \quad (13)$$

Here,

$$\begin{aligned} \dot{r}^{*L} &= \begin{bmatrix} 0 \\ -R \sin(\varphi) \dot{\varphi} \\ R \cos(\varphi) \dot{\varphi} \end{bmatrix} = R \begin{bmatrix} 0 \\ -\sin(\varphi) \\ \cos(\varphi) \end{bmatrix} \dot{\varphi} \\ &= -R \begin{bmatrix} 0 \\ -\sin(\varphi) \\ \cos(\varphi) \end{bmatrix} \begin{bmatrix} 0 & -\sin(\varphi) & \cos(\varphi) \end{bmatrix} \dot{p}_f^L(t) \\ &= \alpha R \begin{bmatrix} 0 & 0 & 0 \\ 0 & \sin(\varphi)^2 & -\sin(\varphi) \cos(\varphi) \\ 0 & -\cos(\varphi) \sin(\varphi) & \cos(\varphi)^2 \end{bmatrix} e^L(t) \\ &= \alpha R M(\varphi) e^L(t), \end{aligned} \quad (14)$$

where $M(\varphi)$ is the corresponding matrix in the third equality, and the second and third equalities come from (10) and (12), respectively. As a result, (13) can be rewritten as

$$\dot{e}^L(t) = -\alpha e^L(t) - \alpha R M(\varphi) e^L(t), \quad e^L(0) = e_0^L. \quad (15)$$

We are now ready to state the following result.

Theorem 1 *Consider the single integrator dynamics of the follower in the leader frame given by (7) and the reference vector r^{L*} defined by (9) and (10), if the follower executes the control algorithm given by (12), then the origin of the closed-loop error dynamics (15) is globally exponentially stable.*

Proof. Consider the Lyapunov function candidate given by

$$V(e^L(t)) = \frac{1}{2} \left(e^L(t) \right)^T e^L(t). \quad (16)$$

Taking the time derivative of (16) along the trajectory of (15) yields

$$\begin{aligned} \dot{V}(\cdot) &= \left(e^L(t) \right)^T \left(-\alpha e^L(t) - \alpha R M(\varphi) e^L(t) \right) \\ &= -\alpha \left(e^L(t) \right)^T e^L(t) - \alpha R \left(e^L(t) \right)^T M(\varphi) e^L(t) \\ &= -\alpha \left(e^L(t) \right)^T e^L(t) - \alpha R \left(e_y^L \sin(\varphi) - e_z^L \cos(\varphi) \right)^2 \\ &\leq -\alpha \left(e^L(t) \right)^T e^L(t) = -2\alpha V(\cdot). \end{aligned} \quad (17)$$

By [34, Theorem 3.6], the origin of the closed-loop error dynamics (15) is globally exponentially stable. $\blacksquare$

Remark 1 Theorem 1 shows that under the proposed control algorithm (12), $p_f^L(t)$ exponentially approaches r^{*L} ; that is, the follower exponentially approaches the desired ring. We further note that when $e^L(t)$ approaches zero, so does $\dot{p}_f^L(t)$. As a consequence, $\dot{\varphi}(t)$ also approaches 0 indicating that $\varphi(t)$ converges to a specific values.

Remark 2 In general, there are many different ways to choose an appropriate rate of change for $\varphi(t)$. In fact, the rate of change given by (10) is derived from choosing $\varphi(t) = \arctan\left(\frac{c_z^L - p_{fz}^L}{c_y^L - p_{fy}^L}\right)$. To elucidate this point, taking the time derivative of the previous expression yields

$$\begin{aligned}\dot{\varphi}(t) &= \frac{(c_y^L - p_{fy}^L)^2}{(c_y^L - p_{fy}^L)^2 + (c_z^L - p_{fz}^L)^2} \left(\frac{\dot{p}_{fy}^L (c_z^L - p_{fz}^L)}{(c_y^L - p_{fy}^L)^2} - \frac{\dot{p}_{fz}^L}{c_y^L - p_{fy}^L} \right) \\ &= \frac{\dot{p}_{fy}^L (c_z^L - p_{fz}^L) - \dot{p}_{fz}^L (c_y^L - p_{fy}^L)}{(c_y^L - p_{fy}^L)^2 + (c_z^L - p_{fz}^L)^2}.\end{aligned}\quad (18)$$

Let $d \triangleq \sqrt{(c_y^L - p_{fy}^L)^2 + (c_z^L - p_{fz}^L)^2}$. In addition, using the fact that $\sin(\varphi) = \frac{c_z^L - p_{fz}^L}{d}$ and $\cos(\varphi) = \frac{c_y^L - p_{fy}^L}{d}$, (18) can be rewritten as

$$\begin{aligned}\dot{\varphi}(t) &= \frac{1}{d} \left(\frac{\dot{p}_{fy}^L (c_z^L - p_{fz}^L)}{d} - \frac{\dot{p}_{fz}^L (c_y^L - p_{fy}^L)}{d} \right) \\ &= \frac{1}{d} \left(\dot{p}_{fy}^L \sin(\varphi) - \dot{p}_{fz}^L \cos(\varphi) \right).\end{aligned}\quad (19)$$

To avoid singularity, the coefficient $\frac{1}{d}$ is dropped from (19), hence (10) is obtained. We further note that if one simply chooses $\dot{\varphi}(t) = 0$, then r^{*L} is just a specific point as in the standard formation control problem.

IV. The Potential Field Method

A. Proposed Control Algorithm

In this section, we propose a different control algorithm for ring formation maneuver. In particular, we first define objective functions, then construct potential functions and potential fields for the follower to achieve the objectives in the leader frame. Specifically, define the following objective functions

$$f_1(p_f^L) = (p_{fx}^L - c_x^L)^2, \quad (20)$$

$$f_2(p_f^L) = (p_{fy}^L - c_y^L)^2 + (p_{fz}^L - c_z^L)^2 - R^2, \quad (21)$$

where $R \in \mathbb{R}_+$ is the radius of the ring. At steady state, when (8) is satisfied, the above objective functions approach zeros implying that the follower is on the desired ring. In other words, r^{*L} is implicitly defined via $f_1(p_f^L) = f_2(p_f^L) = 0$. Next, we construct the potential functions as

$$P_1(p_f^L) = \frac{1}{2} \ln \left(\cosh \left(f_1(p_f^L) \right) \right), \quad (22)$$

$$P_2(p_f^L) = \frac{1}{2} \ln \left(\cosh \left(f_2(p_f^L) \right) \right). \quad (23)$$

Note that $P_i(p_f^L) \geq 0$ for $i = 1, 2$, and $P_i(p_f^L) = 0$ only if $\cosh \left(f_i(p_f^L) \right) = 1$; that is, only if $f_i(p_f^L) = 0$. Therefore, $P_i(p_f^L)$ is a positive definite function of $f_i(p_f^L)$ for $i = 1, 2$, respectively. As a consequence, $P_i(p_f^L)$ is a positive semi-definite function of p_f^L for $i = 1, 2$, respectively. The readers can refer to Appendix A for illustrations of these

potential functions. Now, taking partial derivative of potential functions with respect to p_f^L yields

$$\begin{aligned}\frac{\partial P_1}{\partial p_f^L} &= \frac{\partial P_1}{\partial f_1} \frac{\partial f_1}{\partial p_f^L} = \frac{1}{2} \tanh(f_1(p_f^L)) \frac{\partial f_1}{\partial p_f^L} \\ &= \tanh(f_1(p_f^L)) \begin{bmatrix} (p_{f_x}^L - c_x^L) & 0 & 0 \end{bmatrix},\end{aligned}\quad (24)$$

$$\begin{aligned}\frac{\partial P_2}{\partial p_f^L} &= \frac{\partial P_2}{\partial f_2} \frac{\partial f_2}{\partial p_f^L} = \frac{1}{2} \tanh(f_2(p_f^L)) \frac{\partial f_2}{\partial p_f^L} \\ &= \tanh(f_2(p_f^L)) \begin{bmatrix} 0 & (p_{f_y}^L - c_y^L) & (p_{f_z}^L - c_z^L) \end{bmatrix}.\end{aligned}\quad (25)$$

At this point, we are ready to construct the proposed control algorithm in the leader frame as follows:

$$\begin{aligned}u^L(t) &= - \left(\left(\frac{\partial P_1}{\partial p_f^L} \right)^T + \left(\frac{\partial P_2}{\partial p_f^L} \right)^T \right) \\ &= - \begin{bmatrix} \tanh(f_1(p_f^L)) (p_{f_x}^L - c_x^L) \\ \tanh(f_2(p_f^L)) (p_{f_y}^L - c_y^L) \\ \tanh(f_2(p_f^L)) (p_{f_z}^L - c_z^L) \end{bmatrix}.\end{aligned}\quad (26)$$

B. Stability Analysis

The following theorem presents the result of this section.

Theorem 2 *Consider the single integrator dynamics of the follower in the leader frame given by (7), if the follower executes the control algorithm given by (26), then the follower either approaches the center of the ring or the ring.*

Proof. The stability of the proposed algorithm (26) is analyzed by considering the following Lyapunov-like function

$$V(p_f^L) = P_1(p_f^L) + P_2(p_f^L), \quad (27)$$

and then taking its time derivative along the trajectory of (26) to obtain

$$\begin{aligned}\dot{V}(\cdot) &= \frac{\partial V}{\partial p_f^L} \frac{\partial p_f^L}{\partial t} \\ &= - \left(\left(\frac{\partial P_1}{\partial p_f^L} \right) + \left(\frac{\partial P_2}{\partial p_f^L} \right) \right) \left(\left(\frac{\partial P_1}{\partial p_f^L} \right)^T + \left(\frac{\partial P_2}{\partial p_f^L} \right)^T \right) \\ &= - \left\| \left(\frac{\partial P_1}{\partial p_f^L} \right) + \left(\frac{\partial P_2}{\partial p_f^L} \right) \right\|_2^2 \leq 0.\end{aligned}\quad (28)$$

Consider the set $\Omega \triangleq \{p_f \in \mathbb{R}^3 : V(p_f^L) \leq d\}$ where d is a positive constant. If $p_f^L(0) \in \Omega$, it follows from (28) that $p_f^L(t)$ remains in Ω for all $t > 0$. Thus, Ω is a compact set. By LaSalle's invariance principle (see, for example [34, Chapter 3.4]), every solution starting in Ω approaches the largest invariant set in $E = \{p_f^L \in \mathbb{R}^3 : \dot{V} = 0\}$. Note that (28) indicates $\dot{V} = 0$ only when $p_f^L(t) = 0$; that is, (i) $\tanh(f_1(p_f^L)) (p_{f_x}^L - c_x^L) = 0$, (ii) $\tanh(f_2(p_f^L)) (p_{f_y}^L - c_y^L) = 0$ and (iii) $\tanh(f_2(p_f^L)) (p_{f_z}^L - c_z^L) = 0$ simultaneously. In other words, $E = c^L \cup \{p_f^L \in \mathbb{R}^3 : f_1(p_f^L) = f_2(p_f^L) = 0\}$ which is the center of the ring and the ring. Since both component sets of E are invariant, the largest invariant set in E is itself. In addition, $V(p_f^L)$ is radially unbounded and hence all trajectories will converge to either the center of the ring or the ring. ■

Remark 3 Note that when $p_f^L = c^L$, $V(p_f^L) = V(c^L) = \frac{1}{2} \ln(\cosh(-R^2))$ while when $p_f^L = r^{*L}$, $V(p_f^L) = V(r^{*L}) = 0$. Therefore, for $d < \frac{1}{2} \ln(\cosh(-R^2))$, any trajectory in $\Omega = \{p_f \in \mathbb{R}^3 : V(p_f^L) < d\}$ is guaranteed to converge to the ring. One can observe that the center of the ring c^L is an unstable equilibrium point, and hence, any perturbation about this point will make p_f^L move away from it and converge to the ring. In addition, the condition for the follower to converges to the center of the ring is that the follower is initially located at $[p_{fx}^L \ c_{fy}^L \ c_{fz}^L]^T$ for some arbitrary $p_{fx}^L \in \mathbb{R}$ (i.e., initially, the y- and z-coordinates of the follower coincide with the y- and z-coordinates of the center of the ring) and assume that there is no external force acting on the point mass vehicle. Yet, in practice, external disturbances such as winds, will make the follower converge to the ring.

V. Discussion on the Implementation and the Two Methods

In this section, we discuss the implementation of the methods in inertial frame and the two methods. Specifically, as mentioned in Section II, once the control algorithm $\dot{p}_f^L(t) = u^L(t)$ is obtained in the leader frame, one can substitute it into (5) for implementing on the follower in the inertial frame. In order to implement (5), the follower needs to know the leader's position $p_l(t)$ (for constructing $p_f^L(t)$), velocity $\dot{p}_l(t)$ as well as the flight path angle γ , the course angle χ and their rates (for constructing $\mathcal{R}_L^i(\gamma, \chi)$ and $\dot{\mathcal{R}}_L^i(\gamma, \chi)$). This can be a challenge if the leader is not a cooperative agents or if the communication is lost during the mission. Fortunately, if the follower knows the leader's position $p_l(t)$ and assume that the sampling time is sufficiently small, all other information can be inferred. More details, the leader's position $p_l(t)$ can be passed through a command filter (for example, see [35]) to obtain $\dot{p}_l(t)$. Subsequently, provided that the inertial frame use the North-East-Down (NED) convention, then the flight path angle can be obtained

by $\gamma = \arctan\left(\frac{-\dot{p}_{lz}}{\sqrt{\dot{p}_{lx}^2 + \dot{p}_{ly}^2}}\right)$ and the course angle is given by $\chi = \arctan\left(\frac{\dot{p}_{ly}}{\dot{p}_{lx}}\right)$. Once again, angle rates can be acquired by

passing γ and χ through command filters. Another common scenario is that there is no information exchange between the leader and the follower, yet the follower has the ability to measure the range vector between the two vehicles; that is, the position of the leader in the follower frame $p_l^f(t) = \mathcal{R}_f^L(\gamma_f, \chi_f)((p_l(t) - p_f))$ is given, where γ_f and χ_f are the follower's flight path angle and course angle, respectively. In this case, it is reasonable to assume that the follower knows its own position and angles, then the leader position $p_l(t)$ in the inertial frame can be calculated and other information can be inferred as in the aforementioned procedure.

The two proposed methods are easy to implement and can be further extended in many directions. For instance, the setup of the explicit method is intuitive and simple. In fact, the proposed control algorithm (12) is just a proportional controller. For better performance under disturbances, it is possible to extended the control algorithm to proportional-integral (PI) controller. This would be considered in a future research. Yet, the explicit method seems to be applicable only to this specific ring formation problem. Regarding this matter, the potential field method can capture more scenarios in general. Notably, an arbitrary shape can be achieved as long as they can be defined in term of objective functions appropriately. Furthermore, other tasks including collision avoidance and/or connectivity maintenance can be added to the control algorithm via objective functions. However, one should be careful in both design and analysis stages as when the number of objective functions increases, it is possible for the vehicles to be stuck in some unexpected local minima.

VI. Numerical Examples

In this section, we present numerical examples to illustrate the results of Sections III and IV. For this purpose, we consider the leader's and follower's initial positions are at $[-300; -500; 1000]$ and $[-360; -460; 1100]$, respectively. The desired ring is chosen to have the radius $R = 10$ m and the center is located at $c^L = [-10; 0; 0]$. The leader is commanded to travel at a constant speed of 15 m/s and to track a sinusoidal trajectory on the x-z plane depicted by $\gamma(t) = \cos(0.5t + 0.5\pi)$ and $\chi(t) = 0$. The follower has the initial speed of 12 m/s and implemented the proposed control algorithms (12) and (26), respectively. Figures 2 and 3 depicts the performance of the followers under the control algorithm (12). Specifically, Figure 2 demonstrates the follower's ability to approach the desired ring. In addition, Figure 3 shows that the error $e^L(t)$ defined in (11) quickly converges to a small neighborhood of zero. On the other hands, Figures 4 and 5 depicts the performance of the followers under the control algorithm (26). Particularly, Figure 4 shows that the follower is able to approach the desired ring. Furthermore, Figure 5 presents that the evolution of the objective functions over time. It can be seen that these objective functions also converge to a small neighborhood of zero as expected. Therefore, these numerical examples illustrate the efficacy of the proposed control algorithms.

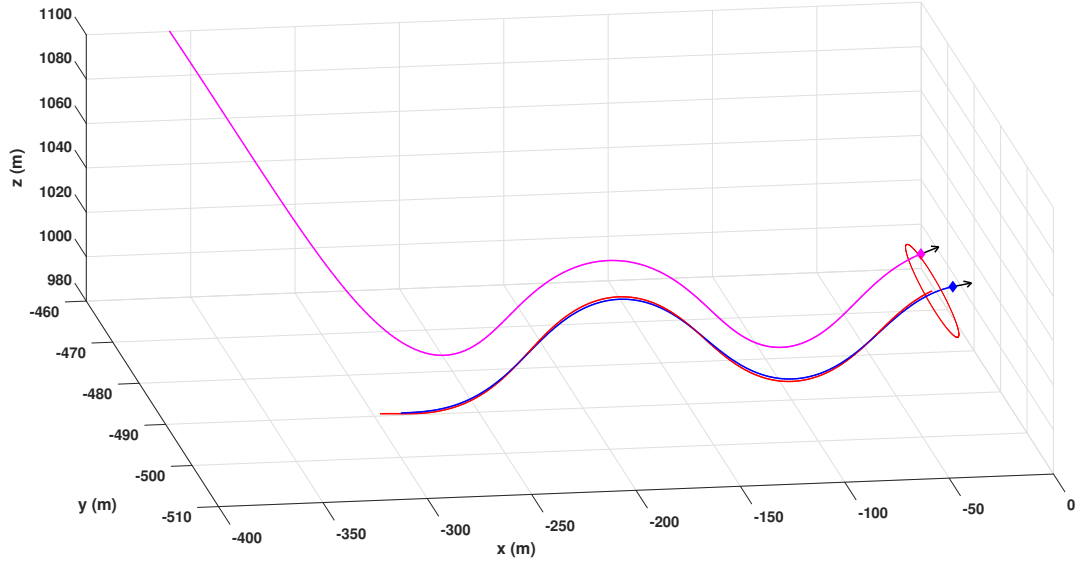


Fig. 2 The evolution of the follower under the proposed algorithm (12), where the blue line denotes the trajectory of the leader, the magenta line denotes the trajectory of the follower, the arrows denote the current velocities of the vehicles, and the tail of the arrows denote the current positions of the vehicles.

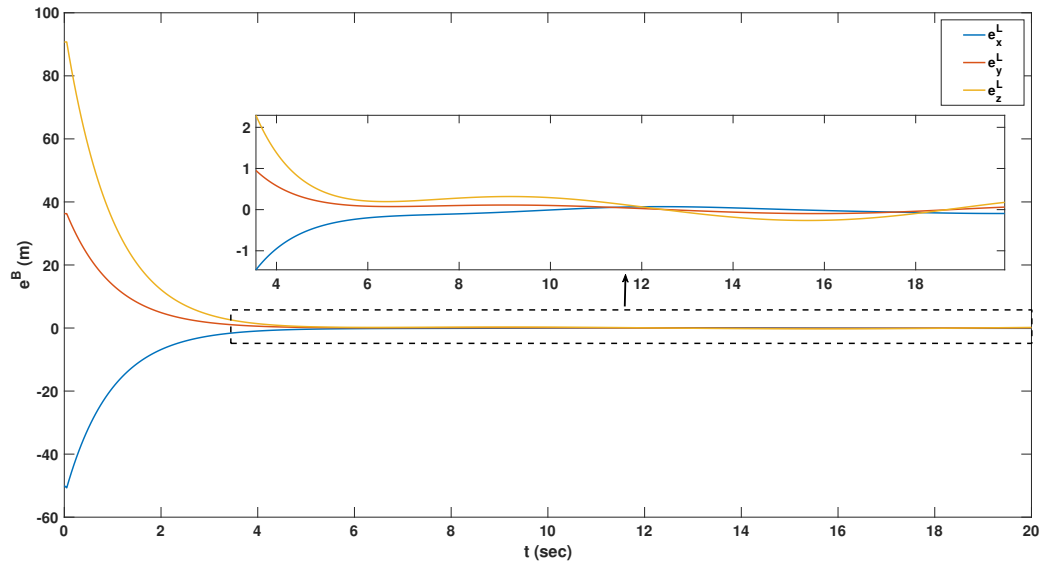


Fig. 3 The evolution of the error $e^L(t)$ defined in (11).

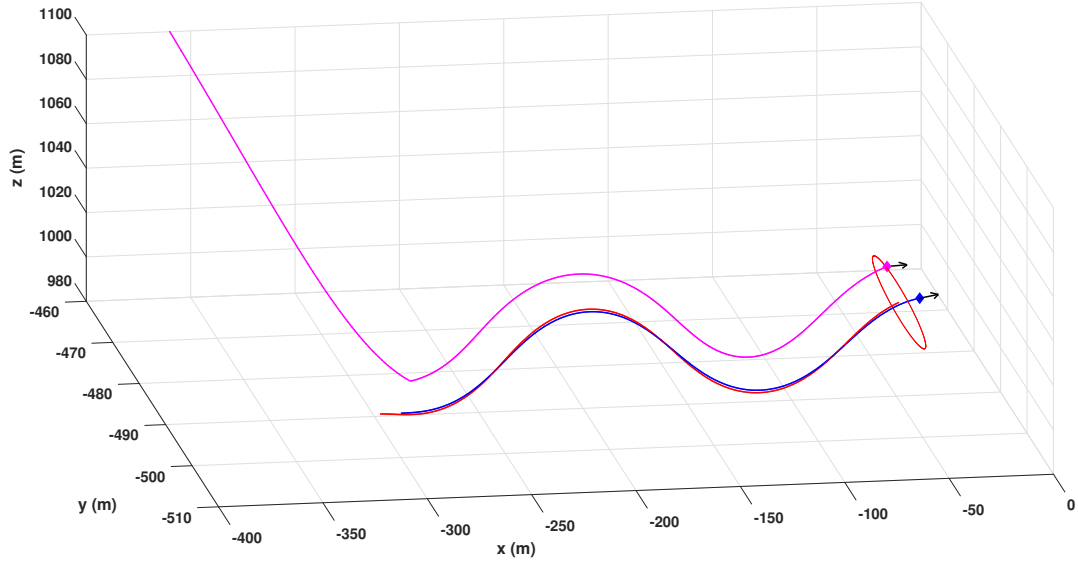


Fig. 4 The evolution of the follower under the proposed algorithm (26), where the blue line denotes the trajectory of the leader, the magenta line denotes the trajectory of the follower, the arrows denote the current velocities of the vehicles, and the tail of the arrows denote the current positions of the vehicles.

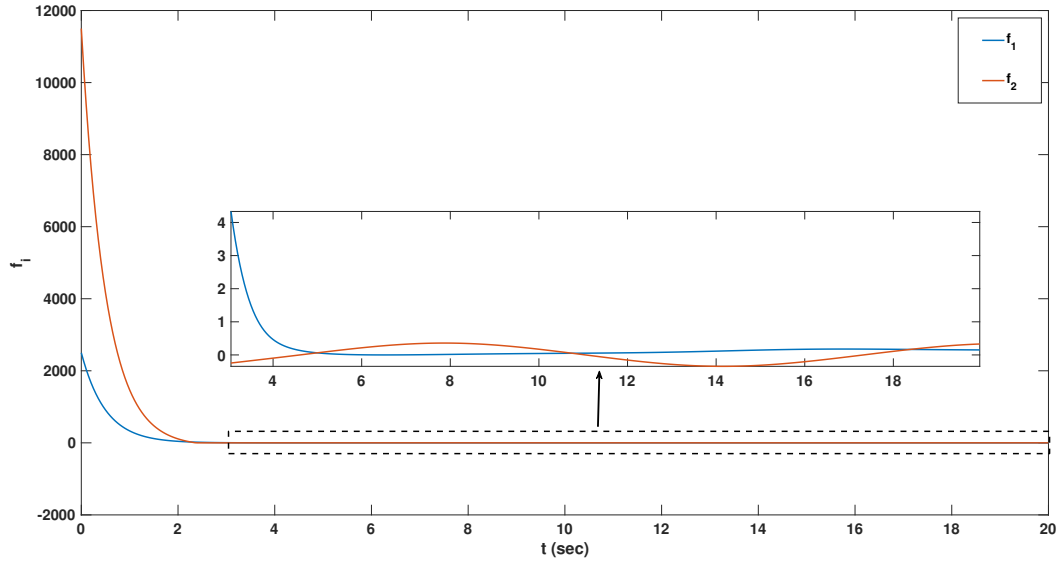


Fig. 5 The evolution of the objective functions $f_1(p_f^L)$, $f_2(p_f^L)$ defined in (20) and (21).

VII. Conclusion

The paper considered the formation control problem, where the desired position of each vehicle in the formation is a set of points (i.e., the ring). Two control algorithms were proposed. The first one explicit defined a point on the ring, which changed depending on the current position of the follower. The second one utilized the potential field as a tool to

define the ring. The stability of the overall system under the proposed control algorithms was analyzed using Lyapunov theory, and numerical examples were presented for justification purposes. Our future works will consider extending the agents' dynamics to higher order and implementing collision avoidance.

Appendix

A. Potential Functions

In this appendix, we illustrate the potential functions depicted in (22) and (23).

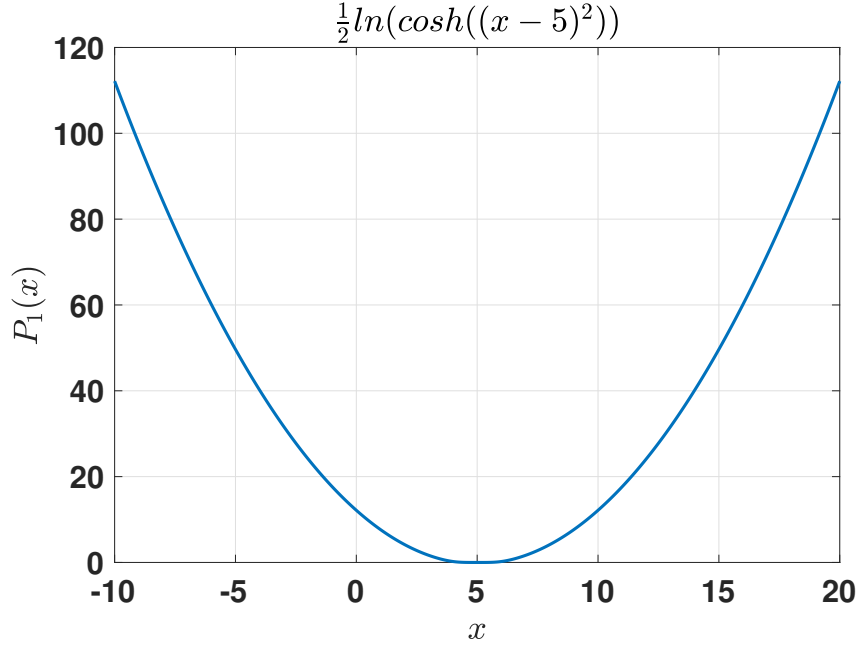


Fig. 6 An illustration of the potential function (22).

References

- [1] Bajec, I. L., and Heppner, F. H., "Organized Flight in Birds," *Animal Behaviour*, Vol. 78, No. 4, 2009, pp. 777–789.
- [2] Lissaman, P. B. S., and Shollenberger, C. A., "Formation Flight of Birds," *Science*, Vol. 168, No. 3934, 1970, pp. 1003–1005.
- [3] Uhlenbroek, C., *Animal life: secrets of the animal world revealed*, Dorling Kindersley, 2012.
- [4] Blake, W. B., and Gingras, D. R., "Comparison of Predicted and Measured Formation Flight Interference Effects," *Journal of Aircraft*, Vol. 41, No. 2, 2004, pp. 201–207.
- [5] Brodecki, M., and Subbarao, K., "Autonomous Formation Flight Control System Using In-Flight Sweet-Spot Estimation," *Journal of Propulsion and Power*, Vol. 31, No. 5, 2015, pp. 1083–1096. <https://doi.org/10.2514/1.G000220>.
- [6] Korkischko, I., and Konrath, R., "Formation Flight of Low-Aspect-Ratio Wings at Low Reynolds Number," *Journal of Aircraft*, Vol. 54, No. 3, 2017, pp. 1025–1034.
- [7] Okolo, W., Dogan, A., and Blake, W., "Effect of Trail Aircraft Trim on Optimum Location in Formation Flight," *Journal of Aircraft*, Vol. 52, No. 4, 2015, pp. 1201–1213. <https://doi.org/10.2514/1.C032865>.
- [8] Oh, K. K., Park, M. C., and Ahn, H. S., "A Survey of Multi-agent Formation Control," *Automatica*, Vol. 53, 2015, pp. 424–440. <https://doi.org/10.1016/j.automatica.2014.10.022>.
- [9] Mesbahi, M., and Egerstedt, M., *Graph Theoretic Methods in Multiagent Networks*, Princeton University Press, 2010.

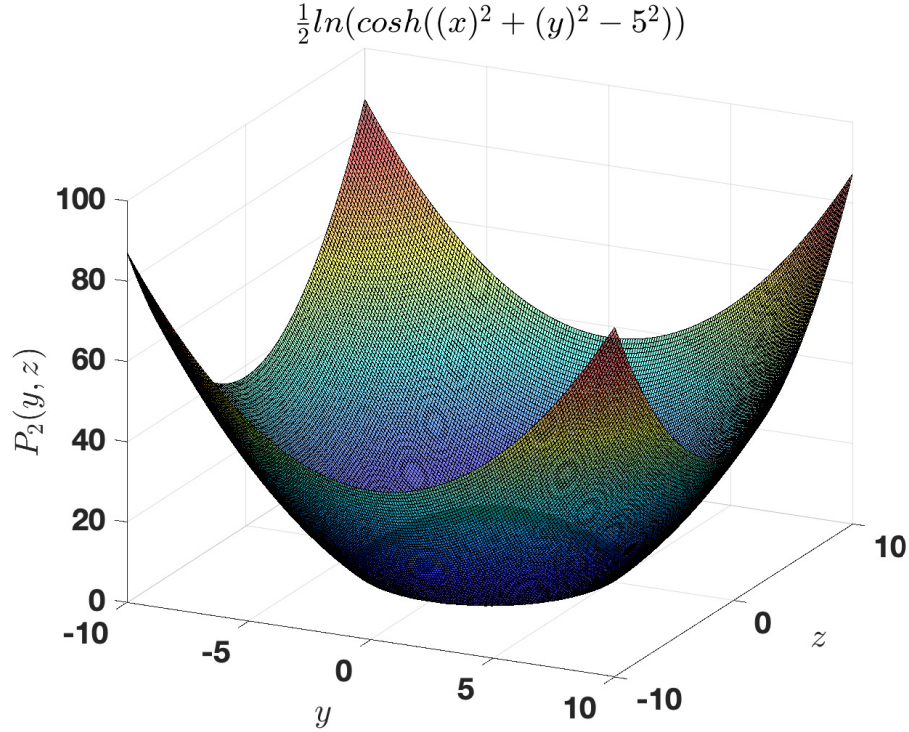


Fig. 7 An illustration of the potential function (23), where the local minima depict the desired ring.

- [10] Ren, W., and Beard, R. W., *Distributed Consensus in Multi-vehicle Cooperative Control*, Springer, 2008.
- [11] Olfati-Saber, R., and Murray, R. M., “Distributed Cooperative Control of Multiple Vehicle Formations Using Structural Potential Functions,” *IFAC Proceedings Volumes*, Vol. 35, No. 1, 2002, pp. 495–500. <https://doi.org/10.3182/20020721-6-ES-1901.00244>.
- [12] Hou, S. P., Cheah, C. C., and Slotine, J. J., “Dynamic Region Following Formation Control for a Swarm of Robots,” *IEEE International Conference on Robotics and Automation*, 2009, pp. 1929–1934. <https://doi.org/10.1109/ROBOT.2009.5152422>.
- [13] Xiao, F., Wang, L., Chen, J., and Gao, Y., “Finite-time Formation Control for Multi-agent Systems,” *Automatica*, Vol. 45, No. 11, 2009, pp. 2605–2611. <https://doi.org/10.1016/j.automatica.2009.07.012>.
- [14] Tran, D., Yucelen, T., and Pasiliao, E. L., “Formation Control with Multiplex Information Networks,” *IEEE Transactions on Control Systems Technology*, Vol. 28, No. 2, 2020, pp. 462–476. <https://doi.org/10.1109/TCST.2018.2884234>.
- [15] Baillieul, J., and Suri, A., “Information Patterns and Hedging Brockett’s Theorem in Controlling Vehicle Formations,” *IEEE Conference on Decision and Control*, Vol. 1, No. December, 2003, pp. 556–563. <https://doi.org/10.1109/CDC.2003.1272622>.
- [16] Cui, N., Wei, C., Guo, J., and Zhao, B., “Research on Missile Formation Control System,” *IEEE International Conference on Mechatronics and Automation*, , No. 1, 2009, pp. 4197–4202. <https://doi.org/10.1109/ICMA.2009.5246527>.
- [17] Ross, S. M., Pachter, M., Jacques, D. R., Kish, B. A., and Millman, D. R., “Autonomous aerial refueling based on the tanker reference frame,” *IEEE Aerospace Conference*, IEEE, 2006, pp. 22–pp. <https://doi.org/10.1109/AERO.2006.1656016>.
- [18] Anderson, R. P., and Milutinović, D., “Stochastic optimal enhancement of distributed formation control using Kalman smoothers,” *Robotica*, Vol. 32, No. 2, 2014, pp. 305–324. <https://doi.org/10.1017/S0263574714000022>.
- [19] Buzogany, L., Pachter, M., and D’Azzo, J., “Automated Control of Aircraft in Formation Flight,” *Guidance, Navigation and Control Conference*, American Institute of Aeronautics and Astronautics, 1993, pp. 1349–1370.
- [20] Casbeer, D. W., Rasmussen, S., Kingston, D., Baker, D., and Garcia, E., “Implementation of Leader-Wingman Flight Formation With Sampled-Data Controller,” *AIAA Scitech 2019 Forum*, , No. January, 2019.

- [21] Dargan, J., Pachter, M., and D'Azzo, J., "Automatic Formation Flight Control," *Guidance, Navigation and Control Conference*, American Institute of Aeronautics and Astronautics, 1992.
- [22] Lavretsky, E., "F/a-18 Autonomous Formation Flight Control System Design," *AIAA Guidance, Navigation, and Control Conference and Exhibit*, American Institute of Aeronautics and Astronautics, 2002.
- [23] Delin Luo, Wenlong Xu, Shunxiang Wu, and Youping Ma, "UAV Formation Flight Control and Formation Switch Strategy," *2013 8th International Conference on Computer Science & Education*, IEEE, 2013, pp. 264–269.
- [24] Pachter, M., D'Azzo, J. J., and Dargan, J. L., "Automatic Formation Flight Control," *Journal of Guidance, Control, and Dynamics*, Vol. 17, No. 6, 1994, pp. 1380–1383.
- [25] Pachter, M., D'Azzo, J. J., and Proud, A. W., "Tight Formation Flight Control," *Journal of Guidance, Control, and Dynamics*, Vol. 24, No. 2, 2001, pp. 246–254.
- [26] Binetti, P., Ariyur, K. B., Krstic, M., and Bernelli, F., "Formation Flight Optimization Using Extremum Seeking Feedback," *Journal of Guidance, Control, and Dynamics*, Vol. 26, No. 1, 2003, pp. 132–142.
- [27] Pollini, L., Innocenti, M., and Giulietti, F., "Sensorless Formation Flight," *AIAA Guidance, Navigation, and Control Conference and Exhibit*, American Institute of Aeronautics and Astronautics, 2001.
- [28] Reyna, V., Pachter, M., and D'Azzo, J., "Formation Flight Control Automation," *Guidance, Navigation, and Control Conference*, American Institute of Aeronautics and Astronautics, 1994, pp. 1379–1404.
- [29] Rife, J., "Collaborative Positioning for Formation Flight of Cargo Aircraft," *Journal of Guidance, Control, and Dynamics*, Vol. 36, No. 1, 2013, pp. 304–307.
- [30] Tahk, M.-J., Park, C.-S., and Ryoo, C.-K., "Line-Of-Sight Guidance Laws for Formation Flight," *Journal of Guidance, Control, and Dynamics*, Vol. 28, No. 4, 2005, pp. 708–716.
- [31] Reyna, V. P., "Automation of Formation Flight Control," Ph.D. thesis, Air Force Institute of Technology, 1994.
- [32] Cheah, C. C., "Region Reaching Control of Robots with Motion Constraints," *2008 10th International Conference on Control, Automation, Robotics and Vision*, , No. December, 2008, pp. 1752–1757. <https://doi.org/10.1109/ICARCV.2008.4795792>.
- [33] Bennet, D. J., McInnes, C. R., Suzuki, M., and Uchiyama, K., "Autonomous Three-Dimensional Formation Flight for a Swarm of Unmanned Aerial Vehicles," *Journal of Guidance, Control, and Dynamics*, Vol. 34, No. 6, 2011, pp. 1899–1908. <https://doi.org/10.2514/1.53931>.
- [34] Khalil, H. K., *Nonlinear Control*, Pearson Higher Ed, 2014.
- [35] Farrell, J., Sharma, M., and Polycarpou, M., "Backstepping-based Flight Control with Adaptive Function Approximation," *Journal of Guidance, Control, and Dynamics*, Vol. 28, No. 6, 2005, pp. 1089–1102. <https://doi.org/10.2514/1.13030>.